

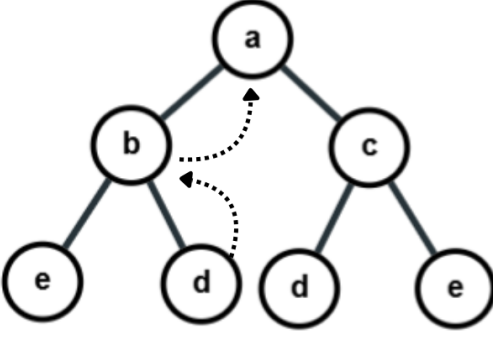
MCQs & SHORT QUESTIONS [6 MARKS]

Q1. [3 Marks] You are given a binary tree with a special property where leaves are represented by 'L' and internal nodes by 'I'. The preorder traversal of this tree is ILILL . Draw the binary tree .	Q2. [1 Mark] A binary tree has 15 nodes . What is the minimum possible height? A) 3 B) 4 C) 5 D) 6
	Q3. [1 Mark] A full binary tree has 7 internal nodes . How many leaf nodes does it have? A) 6 B) 7 C) 8 D) 14
	Q4. [1 Mark] Which of the following statements is true for a complete binary tree? A) All nodes must have two children B) All levels must be fully filled C) All levels are filled except possibly the last, and the last level is filled from left to right D) Every internal node has exactly two children, and all leaf nodes are at the same level

Q5. [12 Marks] You are given the root of a binary tree where each node contains a letter from 'a' to 'z'. Write a function/method named **smallestFromLeaf(root)** that prints the **lexicographically smallest string** among all leaf-to-root paths.

NOTE: You may use multiple helper methods/functions to solve this problem. However, you are not allowed to use any **global or static variables**. Assume that the **BTNode** class is already implemented with the instance variables **elem (char)**, **left (BTNode)**, and **right (BTNode)**.

You cannot use any extra data structures to solve this problem.

Sample Given Tree	Sample Output
	Smallest string from leaf: dba
	Explanation The tree has four leaf-to-root paths: • $e \rightarrow b \rightarrow a = \text{"eba"}$ • $d \rightarrow b \rightarrow a = \text{"dba"}$ • $d \rightarrow c \rightarrow a = \text{"dca"}$ • $e \rightarrow c \rightarrow a = \text{"eca"}$ ✓ Compare the strings: "eba", "dba", "dca", "eca". Strings starting with 'd' are smaller than 'e'. So, compare "dba" and "dca". ✓ For second letter: 'b' < 'c' $\rightarrow$ "dba" is smaller. Therefore, the smallest string is "dba".
Python Function Signature	Java Function Signature
<pre>def smallestFromLeaf(root): # Write your code here</pre>	<pre>static void smallestFromLeaf (BTNode root) { // Write your code here }</pre>

** **Lexicographical order:** Compare strings from left to right. The first different character decides which one is smaller. If all characters are the same, the shorter string is considered smaller. Example: "abb" < "abd" and "ab" < "aba".*

